# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME: Cryptography and Network Security**
**COURSE CODE: CSA51**

**COURSE OUTCOME COVERED**

**CO1:** Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

**QUESTIONS: 5** **Total Marks:50**

Annexure A

## APPLICATION- BASED QUESTIONS

***Code of Conduct:***

*I* yamuna.k */***192511013** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

*Yamuna*

***Signature of the student:***

Annexure A

# APPLICATION- BASED QUESTIONS

**1. Scenario: Event Communication Security**
A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

**2. Scenario: Intercepted Suspicious Message** A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

**3. Scenario: Secure Messaging System** A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

**4. Scenario: Suspicious Digital Asset Investigation**
During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

**5. Scenario: Legacy Encryption in Corporate Communication** A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: 'WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4- Exemplary | Level 3- Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Understanding of Concepts | 25 | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design | 30 | Well-planned, systematic implementation with optimal solution | Correct implementation with minor issues | Basic implementation with errors or inefficiencies | Incorrect or incomplete implementation |
| Use of Tools | 20 | Effective and advanced use of tools/technologies with proper configuration | Appropriate use of tools with correct execution | Limited or inconsistent use of tools | Incorrect or minimal use of tools |

| | | | | | |
|---|---|---|---|---|---|
| Analysis of Results | 15 | Insightful analysis wi'lh accurate interpretation and validation of results | Good analysis with reasonable interpretatio n | Basic analysis with limited insights | Poor or incorrect analysis of results |
| Documentat 10n | 10 | Clear, well-structured documentation with diagrams, code, and results | Organized documentati on with necessary details | Basic documentatio n with m1ss1ng details | Poor or incomplete documentatio n |

# Applications - Based Questions

## $CO_1$ Assessment Tool - 1

Name : K. Yamuna

Reg No : 192511013

Course Code : CSA 5112

Course name : Cryptography and Network Security

positions

I {1 llAf,vJ :

P1rt.u.itcd : M E:f T AT NoCN

p?tu,t ME.E\A-T?IOON

?ie..d tJ 1 1 X £ 'I. l< S s; 'R

f t.1c? ? ; &!llx' fX RsS- R

$plUM CVt-Q ? ? ?hLL-?

f

/

/.

[redacted]

[redacted]

AV'I auttw.t.¥.d. tn.a..ml?o Ll-}<:J Ikc ll 61.wl-l' I) k-l□'- h •l-J, /r, fo

d.AGu.w,.r 1kt C.lph£..□·!..c,d . Ette vi lc"ttl."I [t.1 h 1,i Ll ted {-!ll.U1

**p□Ut,□□** ***bctcku.,1a,lJ.***

[redacted]

[redacted]

[redacted]

[redacted]

**X** **>T**

[redacted]

[redacted]

AnaJtyiLi : fu □ Uf□ GU □ tD □ntVLi aM u

□- tcLJ,u;I . J-1□ , ,u p-tD-lL/dU □ ,1-£,(_1,u..i +V ,,l:,e,c

i"klu ___ OVI,u; **a.r** pW-Mib t.e_ fWVI - .Z..UO A□ **tJ - tnut2{lP**

Lt -t,•t,)) lr1.,;LLf-abtD. □( b),,VU.□ Cl.ct>r\,l LcLl e_M(.U.J□0-l/i k

**PJCt •#14**

**e.DVLf-t tlcvt□**

**bLW-l-1** *towl--l*

2. Frequency Analysis - Simple Substitution Cipher

a) Deduce the most likely original message.

Given ciphertext: GSRH RH Z HVXIVG

The ciphertext is consistent with the Atbash cipher, substitution method in which each letter is replaced by its opposite in the alphabet.

71N- ? is :

,4 L.--? Z

13 L →Y

C ⟷ X

D ⟷ W

E ⟷ V

Applying the same substitution to decrypt the message:

| Ciphertext | GSRH | RH | | HVYIVu, |
|---|---|---|---|---|
| pl?t | TH IS | 1S | A | SECRET |

• (OU-qu.,u:i,I ? ? TH IS tS A S EC F'< E.T

[redacted]

b) Justify the answer using letter patterns and cipher characteristics.

1- IAJ [redacted] -Lei,iq,f-t-i : 'Tke. ti-fkcJr:Urt ,vtGU ·fke palmU1 ,of ¥LI)-/

µ,. [redacted]( rhLw k[redacted]) , [redacted] .lcttL[redacted] a,n£< [redacted]x .ltm.w.

2,, uOwiet[redacted]W-t11iu[redacted]cl-':I : fu 1tLtO - Utlt'1 [redacted]v(O,vti R 1-r wz.ccp1 ro I,s

CtM +Vt{' ,\inqLr. - .lttte,,., wo¼J X [redacted] 1o_A

;,. S[redacted]Wai[redacted]o.0 paJU,,"Lv, : fCt,C,.VI 1e:m,., [redacted] Co.vt,[redacted] mit

[redacted]f>rC--w:Q Jy; -fvLQ. ct/.p,VLa,,[redacted]i .

.If-· V.u Lit etil[redacted]Pn [redacted]AppLL-f W1!j 1W ,41bcwVJ 1'Y/£1.p-rinj 0£jCLUIJ 1D

--rH15 15 A SE C1<£ T up,to[redacted] tt,,,Q. qfvJ2..V1 [redacted]Ud—

t,. H.e.a...i{LWfbU-1 [redacted]t- : fu clu,l,tffted ne..n,iC--llrU [redacted]J

t)Utvvl--maJ-i [redacted] C.o-t,te,d o..nJ. [redacted] context of a

suspicious message.

Analysis : Substitution ciphers preserve word lengths and

le.lw p[redacted] , wWCJIJ (ou,1 [redacted] a;tt[redacted] [redacted]U-ty {w.WIMt VJ

[redacted] known language patterns

th[redacted]

p).a1,vi,tu..t JrvYL-p-u 1,ujnAfu--1Lbn me)Ct,C-GLJ

[ruJr. [redacted] [redacted]

[redacted] h IM!.uJJ [redacted] b-l' c.w.J lo [redacted]d y,_Vl,¼,"tivte [redacted]VI

C[redacted]'[redacted]



3. Vigenere Cipher - keyword - Based Encryption

a) Demonstrate how the encryption process transforms the message step by step.

Given:

plcLtt.t t( t i 1 fl E I L 0

kLt; IUCtd ; kE y

Encryption method : Vigenere cipher

Tkt \.l.l'}tnLU C!_pkV/ we,? 0. <-te.r.c.a..+-ritg kUj UJCU.c/ ,Jo ,leta11,tt./,

tw ?kltt- Clf1'LU:d fo ? plaLvite.,d UJtc,9-1

Step 1: Repeat the keyword

plain text : HELLO

key word : KEYKE

Step 2: Convert letters into numerical values

(.;L\l I+?0, E.::: 1, (_ ' ,'';(= ?)-

ptCtJM,,lex:t Hl1) EC4) l( fJ) LC r1) DC1.1t)

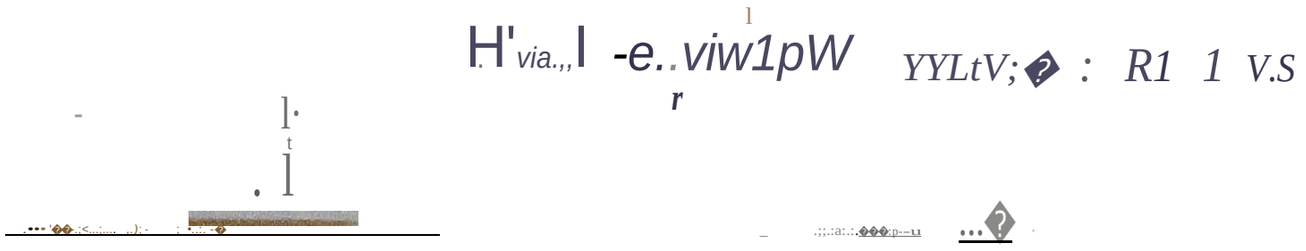
I-(l4,w? k lto) ?Ctt) ?(»t) k (10) E (4)

uJ c.ula.ftp14 t.f ID tt+Jt 11'f.2? t t + l D r4 ?t-1t

?Lt.Li n-u-,d a? li !I 1 ri

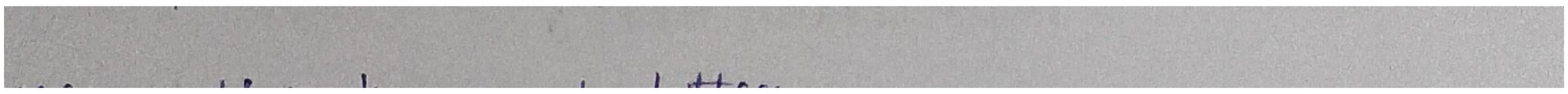
Cif?xJ 1 J V S ?iu;

t f.ffl tt,y,pt-fa? ?vn.cJ.a: Cr= CP; ?? i ) rYlocf 2b



1. Thv f<.Ut u..JtuJ. kf \.f -VJ Aq1ca.ket LLt,Ji,f .LI YVla:I c? q/,,

?ft-J ?t fJL!: plcti.vtUX:i



U() 'v? po? lut-J W{;l,t,). )p_fu, '1

3. Tn.e.. ktl»ui t, E al-id Y Af pi-t.uc.ut )>Kj b, e?l °, -4 a.nd



1-t· /n-e_ kUJ-W(),_ct ? o.,f,¼ flte -f-ti.,Vlcl C - ?,.s0

Ctn.al E ? ? CU:ffl-'M .?, 1? ICQ.t,?t0 ? {t}t-'1

t ??

?..04? fuL;wovl 1.auM p-tt;lGlU/4£ ??vd slu__ftJ

VY!AkWlJ;-/W ? cJ,r ? wip,tQ ? >! fl? a_

? t?-

A? ! 1kt l?vu.,u ?l ? PnuL+l? A-k4{J

r

),v!A,l:e..M of ? ? A? t,.H ?,, ?peoi-? o.

h? kup.v<OAJ CaV.l Mv? .?? f? kLptUto

? ? kltf.•. "-:1? 61D--ttp-N? ? ffurW

J,

? k: ? ftSVi Crot-Lf

Corporat





## 4. Steganography - Hiding Confidential Data in an image

tt) A? .t--uPw ,eyw.,e,Jr cpA-iD--W Ct2u.,ld d.td_a,d presence of

**M,,Llld,e,V) cLaJo w.UkLJIJ ? __;__?**

**1vi:Lt o.dLu.11P n** :

.?.te_q,a_ frl..0-q,LCLf k_,y ;,, 1kc. ic c kvzl q,,u.a ? ,l,Liclfu_g ?t

.l.0,{tP.·LW1.a.,tW 1-1 .ivutC/£ lLVt ?? f,u. , A,t.,lC l-1 ? tl.vJ

.,(__wta.q? ' LtL-t.ctw i;<le b' lL.u:lc.o. lU1.1-Uq_ .e,v,u.ry tuiVJ W ki lh

rvt.a-t? o.. \.n.e½.? L?LQ.. , N?vtcrlLylap? ?tX to

LCI? -1n,.Q b.M 4:.Vb(.Q Dl- 1tu_ ? .

Investigators can use the following techniques:

1, 1'-1? t1vui,? fX'.LLvn. L11te -fc-i.e ..Lvna.q,cJ

?Cl ? Wu,VPBJ ? ? pu.-1-e_J ??

? Mq1M +ta.:t .fn,Q {-Ue. wcw ? .

1. P.le ,0--? Co.wtr,,{l,?V)! ?LQ tCi.e 5.vnaq,e

? ? .tma..q? W ?ti..I-t-j Ll..tw ?e,c.,'Le_J

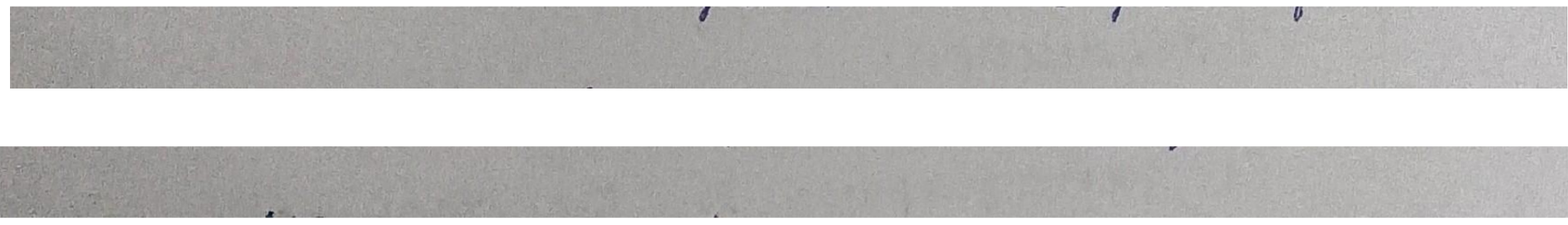
? ? ? /lyt?e. .

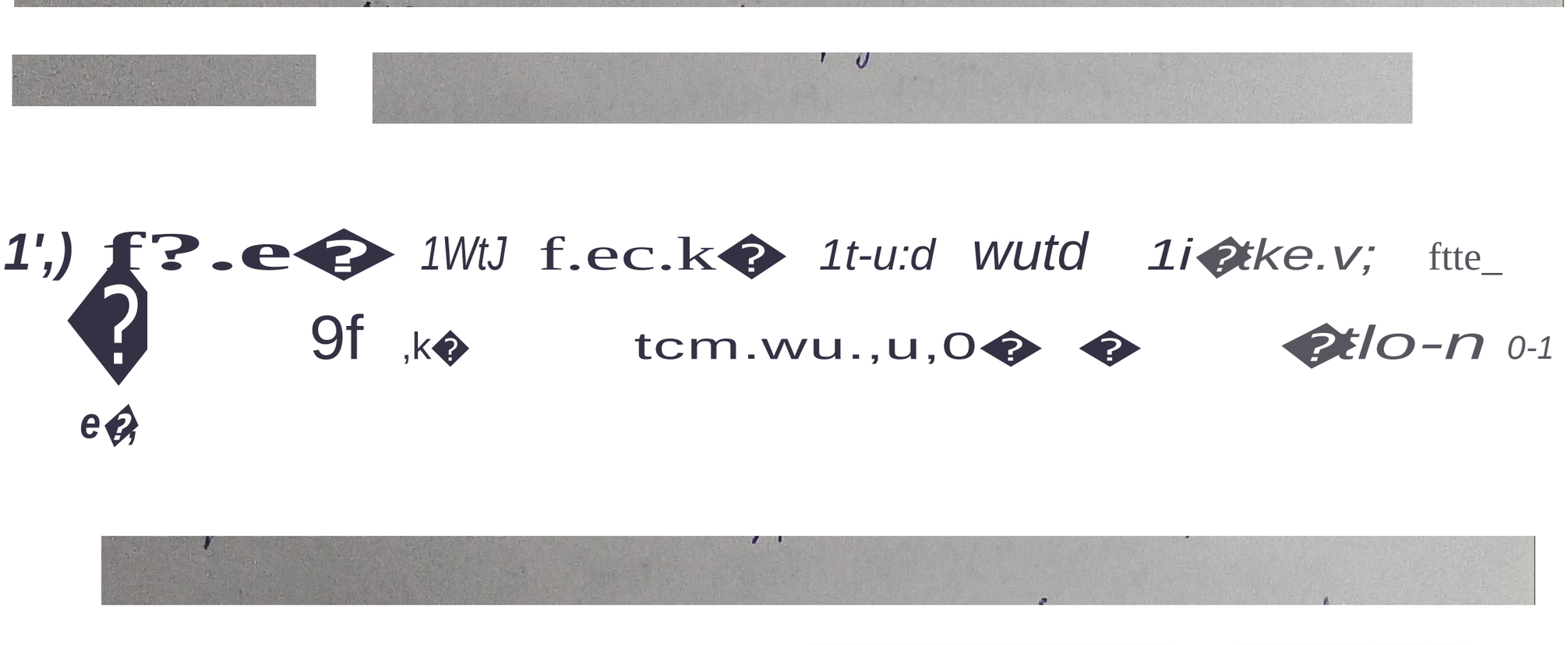
'3• P-L:u.J Cl.vL.a.ly,,'vLl ; ! x_a_VYt..ine P-f¼,.,,( Vo.,lu.e? ?

tn-l(M • ? 4-m t.u--uLt-4.,l pcitrc,.1,? fl-LCd vYi£UJ Jt,,d.Jra.-f.c

;VU?.14?0.

4- Statistical analysis : use steganalysis tools to detect irregularities in pixel distributions or least significant bits (LSBs)

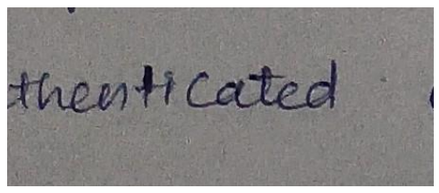








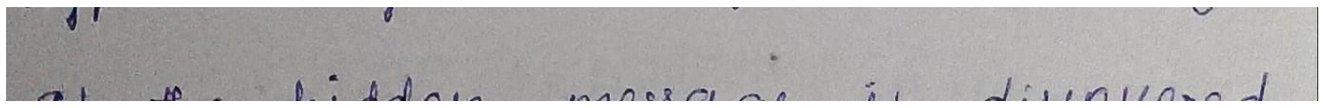












. · l 'f'VIL)(;f_
kiMe.-0
OWt.WLtt..t1-
'ic.a_.tUW] lu.lU
t-
u,t.M'U'ctcd.
. t

5. Transposition cipher - Three - level Arrangement

a) Determine the most probable original message by reversing the encryption process.

Given:

ciphertext : WECRLTEERDSOEEFEAOCAIVDEN

Encryption method : Three - rail Rail Fence Transposition

t?-l£,,.

tJLL?? e{ AC?? : 3

A transposition cipher rearranges the positions of

t ,fr? without changing the letters themselves.

Step 1: Understand fu three - rail arrangement.

In the Rail Fence cipher, the plaintext is written diagonally across three rails in a zigzag pattern.

for example: the first letters of a message follow this pattern:

W

?

F

The letters follow a zigzag pattern through three rows the actual arrangement continues across the complete message.

j
t

Step 2: Reverse the transposition

zigzag pattern using three the letters diagonally in the

Original plaintext:

WEAREDISCOVEREDFLEEATONCE

with spaces added for readability:

WE ARE DISCOVERED FLEE AT ONCE.

Step 3: Verify the result

NECRLTEERDSOEEFEACAIVDEN

This matches the given ciphertext.

b) Critically evaluate why this method is vulnerable in modern communication systems.

1. No letter substitution: The original letters remain

l utters remain th same as in e plaintext

. Predictable st ture : Attackers can test different

,If ? ?/i-uU'L: tth le ter meaningful text Utv1 k) · 1 cilJ--?VLt

,\..eUvl LLVL,b (t,t,tJ ,Lo{)t-, tJ(SL\ /tnJ.UkYILi'l£}.fru ? ?I-.

\ . JAL t P+ VYL? />l,c.,U,,1,-{:.,,.TI.j : ThR.. "Vlitiko.d ? J1.,g,f

pvt-.o.u.l.d:R ,1,? U21,1_...fr detdf a_L1'kJ 0-! ..tvt..,1-vf,ti.:b.j pt? di fJ-0.

C) pi? ? ?ea). ,e,?frnQM.J 1o LWL?

??vttt?

"/k.,.Q.. WVYLJ'.) av? [l-r,C,.u ld- ..,u,,p Laa. --k f2a_d 1-e.t"YC.€. Cif'1.0i

U..,U..AVI A£-r G,C f--1 aLLi iu.vcl-t cd.Lu:l _Lt?,p-H{)..,VI t,ulvig Cl /).{_,C.1./,J,_e.lij

1 {.,flt.Q.,,\cctul kLLj ·

A ES ('1C M f>-'t-C t,,,u;l.e,\ Cc vt-,Fclu,J.·fa..li.:tlj a.vid ..f, vt,ie_q,,i.d fj

? C£tJi.bM• .:;-fAt? „ 1 Vl-L C"' ,,,,.,Dl1 JU,{ ? t --- v A,kc..u_,,L.d o:J)yO iYL11-t£. ct Li...--t. 1 r •-

?/:?·ff(9V) f<_o.,{,1 ?c - ł - O,{.,(...t,'yJ .-fc Ct.LdC1.-t)?e...d

,l.,f/\,'lf>?K_,,,1/5

4 Vl...Cl-UJ,..µ : ?Z),1-i? t\L,iJ2LVL,'>-P?-i it€.CW} c.i-(0i-i.<2.,1 lLL L,tl,,

CU/l-D-fti.,.e,I ? uJ_ t. t,f? ?u, (£1 noA- ?-1,t i?£ 1><---/-1;-f,i cLevd

pw-½c kc{QJ/1 ? /IVI-c_el.,Q,:11/1 o.:tt a.-L /::--)- ;R-e.,p-£cCA.-;:_,,ig 1 l-u 1-?P-L-Lj

. . . ? H,,, a ?t.v',I l<,,/.Id - {A.vi.e..ol R.,v,(.,',,.t.,,J (9ftQ.v, a.f.'3"J-1./1 k

?ko.d U,U,1 v 1

.M ? ,p-'-C0G.-Hca...l AotuJr"[C.VI 1l!11 OJtpa,lA.-1.e. CtJJrL 1 "1"1-LU"IJC..?